NSA Ghidra: Reverse Engineering & Binary Decompilation

The official open-source suite used by intelligence agencies and malware analysts to decompile binaries back into C.

🕵️ 1. What is Ghidra?

Ghidra is a software reverse engineering (SRE) framework developed by the NSA's Research Directorate, released as free open-source software (Apache 2.0).

- **SLED Engine:** Disassembles x86, ARM, MIPS, RISC-V and PowerPC.
- **Decompiler:** Converts raw machine opcodes directly into readable C source code!

🚀 2. Quick Setup in 2 Minutes

```
# 1. Requires OpenJDK 21
sudo apt update && sudo apt install -y openjdk-21-jdk

# 2. Download from official NSA GitHub
wget https://github.com/NationalSecurityAgency/ghidra/releases/download/Ghidra_11.1.2_build/ghidra_11.1.2_PUBLIC_20240709.zip

# 3. Unzip and run GUI
unzip ghidra_*.zip && cd ghidra_*
./ghidraRun
```

⚙️ 3. How to Decompile Any Binary

- **Step 1:** File ➔ New Project ➔ Non-Shared Project.
- **Step 2:** Drag and drop any `.exe`, `.so`, or ELF binary into the project.
- **Step 3:** Double-click the file to open CodeBrowser ➔ Click **Yes** to Auto-Analyze.
- **Step 4:** Check **Decompiler Window** on the right pane: raw assembly is converted to clean C functions!

🔑 4. Key Assembly Cheat Sheet

```
MOV EAX, [EBP+8]   ; Load function argument
CMP EAX, 0x1337    ; Check license key / secret
JE  0x00401200     ; Jump if equal (password correct)
JNE 0x00401250     ; Jump if not equal (access denied)

Patch instruction: Change JNE to NOP or JMP to bypass auth!
```

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Official GitHub: **github.com/shubh74/coderking-studio**